T- testing

Difference of Means for two independent Samples.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{S^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$S = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}}$$

$$t = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\left(\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

degree of freedom d.f. $= \nu = (n_1 + n_2 - 2)$

Q. Samples of two types of electric bulbs were tested for length of life and the following data were obtained

	Size	Mean	SD
sample-1	8	1234 h	36 h
sample-2	7	1036 h	40 h

Is the difference in the means sufficient to warrant that type-1 bulbs are superior to type-2 bulbs?

Solution :- Here $\bar{x}_1 = 1234$ $S_1 = 36$ $n_1 = 8$

$\bar{x}_2 = 1036$ $S_2 = 40$ $n_2 = 7$

$H_0 : \bar{x}_1 = \bar{x}_2$; $H_1 : \bar{x}_1 > \bar{x}_2$

Right tailed test is to be used

Let Level of significance (LOS) :- 5%

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\left(\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{198}{\sqrt{\left(\frac{21568}{13}\right)\left(\frac{1}{8} + \frac{1}{7}\right)}} = \frac{198}{21.0807}$$

$$t_{cal} = 9.39$$

d. f. $v = n_1 + n_2 - 2 = 13$

From table $t_{0.05}(v = 13)$ for one tailed test $= 1.77$

$t_{cal} > t_{tab}$. $\Rightarrow$ Ho is rejected & H_1 is accepted

$\Rightarrow$ type-1 bulbs may be regarded superior to type 2 bulbs at 5% LOS.

Q. The mean height and the SD height of 8 randomly chosen solidiers are 166.9 and 8.29 cm respectively. The corresponding values of 6 randomly chosen sailors are 170.3 and 8.50 cm respectively. Based on this data, can we conclude that soldiers are, in general, shorter than soldiers?

Soln.

$\bar{x}_1 = 166.9 \qquad s_1 = 8.29 \qquad n_1 = 8$

$\bar{x}_2 = 170.3 \qquad s_2 = 8.50 \qquad n_2 = 6$

$H_0 : \bar{x}_1 = \bar{x}_2 \quad ; \quad H_1 : \bar{x}_1 < \bar{x}_2$

Left tailed test is to be used. Let LOS be 5%.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\left(\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{-3.4}{\sqrt{\left(\frac{983.29}{12}\right)\left(\frac{1}{8} + \frac{1}{6}\right)}} = -0.695$$

$$v = n_1 + n_2 - 2$$

$$v = 8 + 6 - 2$$

$$v = 12$$

$$t_{0.05}(v=12) = 1.78$$

$$|t| < t_{0.05}(v=12)$$

$\Rightarrow$ H_0 is accepted and H_1 is rejected.

$\Rightarrow$ Based on the given data, we cannot conclude that soliders are, in general shorter than sailors.